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Defining the Quantitative Criteria for Two Basketball Shooting Techniques

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Abstract: For basketball technique elements to become stable and automated, the technique must be executed correctly from the initial stages of the learning process. The jump shot is considered the most crucial shooting action, equally effective in various on-court situations. Ideal techniques may not unconditionally be the best for every player because they differ in motor and morphological characteristics. One of the shooting methods is shot with one hand from the spot, which, in terms of complexity and required motor abilities, is at a lower level of demand when compared to the jump shot. The first aim of this research is to define the kinematic pattern of the jump shot and the shot with one hand from the spot and determine if there are differences between them. The second aim is to establish whether there is a significant difference in the level of motor skills between the two groups. The participant sample consists of 29 junior basketball players. They were divided into two groups: G1 = players who perform the shot with one hand from the spot (N = 14); G2 = players who execute the jump shot (N = 15). MANOVA was used to determine the differences in observed kinematic parameters between the group and in the measured tests assessing motor abilities. The kinematic parameters of the upper extremities, the parameters of the basketball player's take-off, and the angle of the ball's impact were analyzed. It was determined that the kinematic pattern of these two shooting techniques significantly differs (F = 12.86, p = 0.00). There was a statistically significant difference between the group of participants performing the jump shot and the group performing the one-handed chest shot from the spot (F = 5.51, p = 0.00) in motor ability tests. The group that performed the jump shot showed significantly better results in three of the five measured tests (PLANK, MED_THR, POL_BACK). Shooting with one hand from the spot is a technically correct shooting method recommended to be performed until the player possesses a certain level of motor skills that enables proper execution of the jump shot.

Keywords: jump shot; motor abilities; motor knowledge; kinematic parameters

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1. Introduction

To adopt basketball-specific motor structures, active participation in a prolonged process of learning and practice, i.e., basketball training, is necessary. Under its influence, a basketball player reaches a level where their abilities and knowledge are so developed that they can purposefully control movements during the execution of dynamic stereotypes required in basketball [1]. Only mastery of technique, or specific motor knowledge, will enable a more efficient realization of tactical ideas. Krause, Meyer, and Meyer [2] demonstrate through their research that basic techniques should be learned as early as the ninth or tenth year of age. Improper learning of basketball techniques can lead to the formation of incorrect motor patterns, which are challenging to correct in later stages [3].

The jump shot is the most common way to put the ball into the hoop [4]. Consequently, the jump shot is considered the most crucial shooting action, equally effective in various on-court situations, whether from a stationary position, while in motion, or following a dribble

or a passed ball [5]. The significant advantage and purpose of the jump shot lie in elevating above the defensive player, allowing for the ball to be released into the hoop from a higher position, significantly complicating the activities and effectiveness of defensive players [5]. The jump shot represents a complex motor movement, and the quality of its execution greatly influences the situational efficiency of every basketball player [6]. The execution of a proper jump shot requires less time compared to other shooting techniques [7–9], making it even more purposeful.

Erčulj and Štrumbelj [10] emphasize that the ideal technique may not unconditionally be optimal for every player because they differ in motor, morphological, and anthropometrical characteristics. This realization provides significant opportunities in the context of using various elements of technique to achieve a specific (common) goal [7]. The specific subject of this research refers to shooting from a greater distance, where the technique of individual players varies; that is, it differs so much that it is possible to claim that there are two shooting techniques. One of the shooting methods is the shot with one hand from the spot, which, in terms of complexity and demand, is at a lower level compared to the jump shot. The shot with one hand from the spot is most commonly applied in younger age selections, and one reason for using this shooting technique is the underdevelopment of the musculature of the upper and lower extremities [5]. Furthermore, this is also because, according to biomechanical analysis, this element is closest to the correct execution of the jump shot, and its adoption facilitates an easier transition to the previously mentioned shooting technique, which has become a universal and widely accepted shooting technique [11].

The dynamics and characteristics of today's basketball game also influence player selection in basketball. In the contemporary game, preference is given to tall and powerful players, with an advantage given to athletes who are anthropometrically and constitutionally larger. For instance, Karalejić and Jakovljević [12] investigated differences in the anthropological characteristics of young basketball players aged 12 and 14, as well as the correlation of these characteristics with technical skills. According to their research, differences exist and are significant. As expected, 14-year-old players show significantly higher values in the parameters of longitudinal anthropometric characteristics and significantly better technical skills. In practice, it is often observed that young basketball players incorporate the jump shot into their repertoire of technical elements too early. Specifically, when their motor and anthropometric development is not sufficiently advanced for proper execution, the jump shot is performed with certain deviations. Such execution can significantly disrupt the motor task pattern. If it is not performed correctly, losing its purpose, then it may have serious consequences for the future career of a young basketball player [3]. As mentioned above, players who are not in a satisfactory stage of physical development or simply do not possess the necessary characteristics cannot execute this element. This group includes players from younger age categories, as well as female basketball players. What they share in common is lower level of motor skills, and being anthropometrically smaller. Consequently, the mentioned characteristics result in different values of kinematic and kinetic parameters. Additionally, players from younger age categories are not at the training level necessary for the execution of this element in their phase of a player's career because it has already been established that this element is fundamentally very complex for proper execution [13]. Some authors also emphasize that the execution of the jump shot depends on the player's gaming experience [7,14–16], which further complicates matters for younger basketball players.

The entire basketball game, including shooting technique, can be analyzed from a structural, functional, and biomechanical standpoint, often utilizing biomechanical analysis to investigate basic kinematic and dynamic parameters of movement. Biomechanical analysis provides data that can be compared with the model of an elite basketball player, allowing researchers to assess a player's maximum situational efficiency in the basketball game [17]. Similarly, kinematic analysis allows for a comparison of an athlete's performance in a specific element, or a comparison of elements based on certain kinematic parameters.

Therefore, the aim of this research is to define the kinematic pattern of the jump shot and the shot with one hand from the spot from a stationary position and determine if there are differences between them. The subsequent objective is to establish whether there is a significant difference in the level of motor skills between players using the jump shot and those using the shot with one hand from the spot from a stationary position. The authors hypothesize that significant differences will be identified in the kinematic parameters between the two shooting techniques. Additionally, it is expected that there will be distinctions in motor skills, where players who perform the jump shot will be more successful in motor skill tests.

2. Materials and Methods

Participants: The participant sample consists of 29 junior basketball players (15.88 ± 0.69 years old). All participants compete in the highest tier of the Croatian Basketball Federation (I. Junior League). Additionally, within the past year, none of the participants included in the research have experienced any health or musculoskeletal difficulties that could impact the training process and the results of the study. The participants are divided into two groups: G1 = players who perform the shot with one hand from the spot (N = 14); G2 = players who execute the jump shot (N = 15). Before the measurements began, participants received detailed information about the measurement protocol, the benefits, and risks of the research. Upon arrival at the measurement session, all players signed consent forms for participation. This study has been approved by the Ethical Committee of the University of Zagreb Faculty of Kinesiology and research was in accordance with the Helsinki Declaration (nr. 16/2021).

Variables and equipment: Measurement of kinematic parameters of shooting technique was conducted using the Xsens motion capture system (Xsens, Enschede, The Netherlands). This kinematic system consists of 17 wireless sensors, a base station for signal reception and transmission, and software for recording, storing, and analyzing measured data. The validity of this measurement instrument for measuring body kinematic parameters has been previously established (CMC > 0.96) [18]. Additionally, test–retest reliability (CMC > 0.97) and validity (CMC > 0.91) in measuring segment acceleration in space have been confirmed [19]. In the context of basketball, specifically jump shots, this device has been used to observe angular relationships and define the impact of loads on changes in performance technique [20,21].

For measuring kinematic parameters, such as the entry angle of the ball into the basket and the duration of the shot (moment from ball reception to ball release), the 94Fifty smart basketball was used. The 94Fifty ball (InfoMotion Sports Technologies Inc., Dublin, OH, USA) contains nine accelerometers inside, enabling the detection of force, speed, ball rotation, and ball arc from a 360-degree perspective.

The variable sample consists of results from tests assessing the level of motor skills and kinematic parameters of shooting technique performance. The motor skills profile of the participant groups was measured through the following tests: the plank hold (PLANK), squat jump (SQJ), counter movement jump (CMJ), medicine ball throw from a seated position (MED_THR), and backward agility run (POL_BAC).

In the PLANK test, the maximum time sustained in a static position was observed and expressed in seconds. Timing for the PLANK test was conducted using a stopwatch.

Tests focused on defining explosive strength of the lower extremities (SQJ, CMJ) determined the maximum jump height values expressed in centimeters. For measuring explosive strength of the jumping type (SQJ and CMJ), the Optojump measuring instrument (Microgate, Bolzano, Italy) was used.

In the MED_THR test, the maximum values for medicine ball throwing were determined and expressed in meters. Measurement of results in the MED_THR test utilized a measuring tape with a length of 20 m to record the outcome.

The result in the backward agility run test, which assesses the level of coordination, is expressed in seconds (s). To measure results in the POL_BAC test, WITTY photoelectric cells (Microgate, Bolzano, Italy) were used to register the start and end of the test.

In the execution of the two shooting techniques, the following variables were observed:

- Jump_L—jump length (horizontal component—toward the basket);
- Pelvis_P—the difference in the position of the pelvis from the rest phase to the highest position during the shot (vertical component);
- Hand_rec—the angle in the hand joint at the moment of receiving the ball;
- Sh_shoot—shoulder angle at ball release;
- Hand_H—the maximum height of the hand during releasing of the ball;
- Hand_V—maximum velocity of the hand segment (horizontally);
- Forearm_V—maximum velocity of the forearm (horizontally);
- Upperarm_V—maximum velocity of the upperarm (horizontally);
- EA—entry angle of the ball when approaching the rim;
- PSS—player shooting speed (contact time with the ball).

Protocol of investigation: The measurement begins with defining the anthropometric characteristics necessary for comparing participants and calibrating the measuring device. Subsequently, the participant is outfitted with the kinematic system, including the corresponding shirt and 17 sensors secured with straps. The participant then initiates a standardized warm-up protocol consisting of straight running, lateral running, running drills, and dynamic stretching in place and in motion. At the end of the warm-up, specific warm-up activities with a basketball are performed, including practice shots to acclimate the participant to the added ball from the basketball machine (Dr. Dish) and warm up the shooting motion. Before the actual measurement, a calibration of the kinematic system is conducted, followed by a series of 6 shots. Upon the signal from the measurer, recording is initiated with cameras, the basketball machine, the 94Fifty ball, and the measurement of kinematic parameters. The testing is conducted using two shooting techniques: shot with one hand from the spot (G1) (Figure 1) and jump shot (G2) (Figure 2). Both techniques are performed from a distance for a 3-point shot (6.75 m). Five successful shots for each subject were included in the analysis. Then, the measurement of motor skills begins, following the sequence:

1. PLANK
2. MED_THR
3. SQJ
4. CMJ
5. POL_BAC

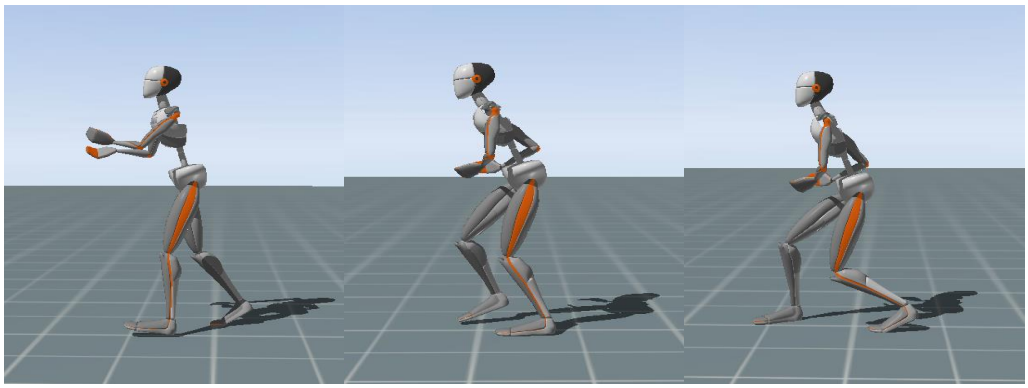

Figure 1. *Cont.*

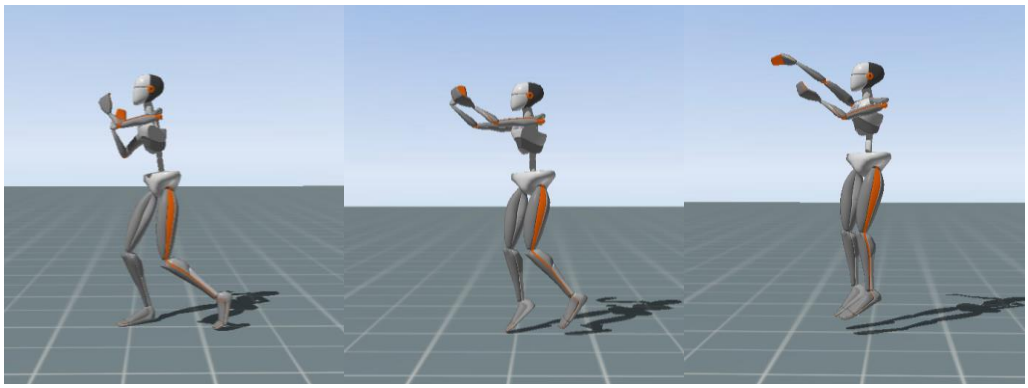

Figure 1. Kinogram of the shot with one hand from the spot performance from Xsens MVN Analyze software 2022.0.2.

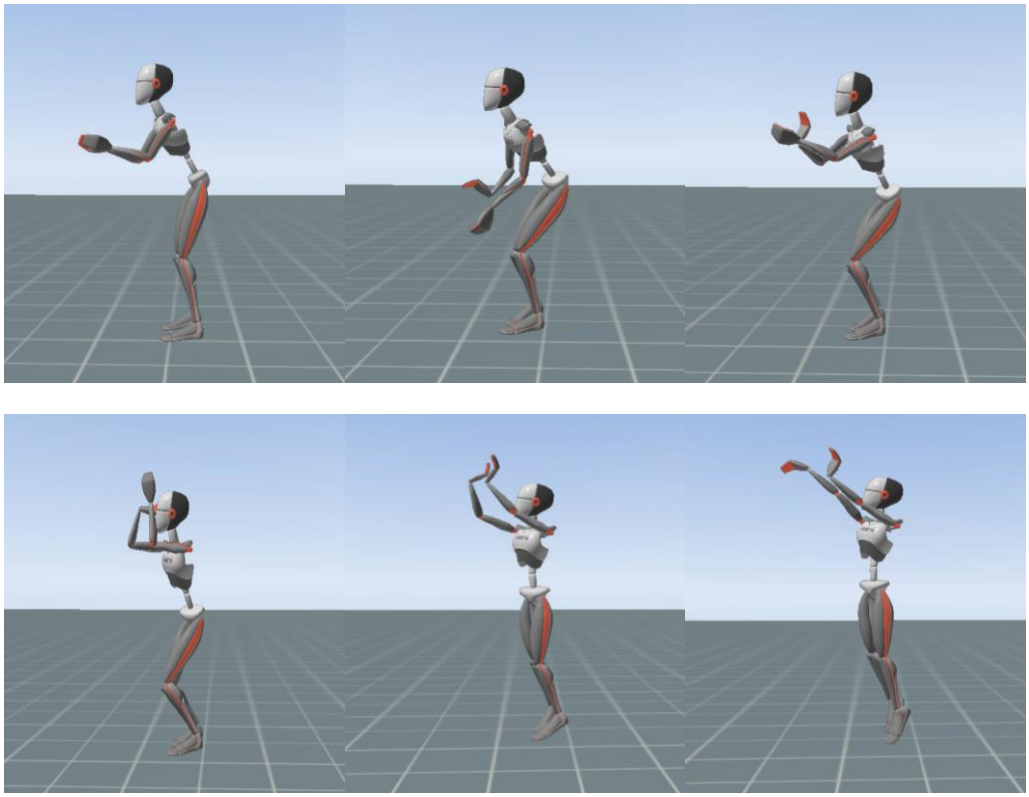

Figure 2. Kinogram of the jump shot performance from Xsens MVN Analyze software 2022.0.2.

With the completion of the motor skills testing, the participant has finished participating in the research.

Classification of Participants into Groups

A team of 5 experts (professors and basketball coaches) decided whether the participant belonged to a jump shot group or a shot with one hand from the spot group based on the initially executed shots.

Statistical analysis: Using G*Power (v.3.1.9.2), the total sample size (N = 90) required for the research was calculated, considering a significance level of $p < 0.05$, a statistical power of 0.8, an effect size of 0.25, and 2 groups of participants (shooters). For statistical data analysis, the software package Statistica v.13.05.0.17 (TIBCO Software Inc., Palo Alto, CA, USA) was employed. Basic descriptive parameters were calculated for all observed variables. The Shapiro–Wilk test was used to assess the normality of the distribution

of observed variables. Multivariate Analysis of Variance (MANOVA) was utilized to determine differences in kinematic parameters between the two observed groups. Similarly, MANOVA was employed to test differences between groups in tests assessing motor skills.

3. Results

A statistically significant difference was found between the two measured groups (F = 12.86, p = 0.00). The following is the whole model for all measured variables during the execution of the two shooting techniques to precisely identify the variables in which the two observed shooting techniques differ.

Table 1 shows a statistically significant difference in all observed kinematic parameters during the execution of shots using different techniques, except for the variable hand height at ball release (Hand_H—F = 2.53, p = 0.11). It is important to note that the group employing the jump shot technique statistically significantly achieved a greater forward jump length during the shot execution (Jump_L—29.18 ± 10.13 vs. 22.01 ± 12.19), as well as a higher jump height (Pelvis_P—32.21 ± 6.19 vs. 30.32 ± 5.01). Additionally, a statistically significantly larger shoulder angle at release was observed in the group using the jump shot technique (Sh_shoot—118.92 ± 15.54 vs. 111.20 ± 12.75). Furthermore, examining the speed of limb segments during the release, the group employing the shot with one hand from the spot technique achieved higher speeds in all observed parameters (Hand_V—4.94 ± 0.88 vs. 4.73 ± 0.71; Forearm_V—5.44 ± 0.79 vs. 4.73 ± 0.71; Upperarm_V—2.71 ± 0.38 vs. 2.46 ± 0.34). However, the shot duration was significantly shorter; i.e., the shot was executed faster, in the group using the jump shot technique (PSS—0.87 ± 0.11 vs. 0.97 ± 0.09).

Table 1. Display of MANOVA for the kinematic parameters of the two observed groups.

	G1	G2		
Variable	**Mean ± SD**	**Mean ± SD**	**F**	***p***
Jump_L (cm)	29.18 ± 10.13	22.01 ± 12.19	14.91	0.00 *
Pelvis_P (cm)	32.21 ± 6.19	30.32 ± 5.01	4.06	0.05 *
Hand_rec (°)	60.62 ± 15.43	51.90 ± 16.65	10.71	0.00 *
Sh_shoot (°)	111.20 ± 12.75	118.92 ± 15.54	10.75	0.00 *
Hand_H (cm)	214.28 ± 11.21	217.02 ± 9.37	2.53	0.11
Hand_V (m/s)	4.94 ± 0.88	4.58 ± 0.59	8.33	0.00 *
Forearm_V (m/s)	5.44 ± 0.79	4.73 ± 0.71	32.07	0.00 *
Upperarm_V (m/s)	2.71 ± 0.38	2.46 ± 0.34	10.12	0.00 *
EA (°)	45.61 ± 4.56	43.31 ± 4.11	16.64	0.00 *
PSS (s)	0.97 ± 0.09	0.87 ± 0.11	33.11	0.00 *

Legend: Jump_L (cm)—jump length (horizontal component—toward the basket); Pelvis_P (cm)— the difference in the position of the pelvis from the rest phase to the highest position during the shot (vertical component); Hand_rec (°)—the angle in the hand joint at the moment of receiving the ball; Sh_shoot (°)—shoulder angle at ball release; Hand_H (cm)—the maximum height of the hand during releasing of the ball; Hand_V (m/s)—maximum velocity of the hand segment (horizontally); Forearm_V (m/s)—maximum velocity of the forearm (horizontally); Upperarm_V (m/s)—maximum velocity of the upperarm (horizontally); EA (°)—entry angle of the ball when approaching the rim; PSS (s)—player shooting speed (contact time with the ball); *—$p < 0.05$.

To determine the differences between the two groups in the measured tests assessing motor abilities, multivariate analysis of variance (MANOVA) was also employed. The results indicate a statistically significant difference between the jump shot group and the shot with one hand from the spot group (F = 5.51, p = 0.00). The overall model differences for each individual motor test and the significance of the differences are presented in Table 2.

Table 2. Results for each variable in the field of motor skills.

	G1	G2		
Variable	**Mean ± SD**	**Mean ± SD**	**F**	***p***
PLANK (s)	125.84 ± 39.18	172.73 ± 39.07	10.40	0.00 *
SQJ (cm)	33.97 ± 5.24	34.41 ± 4.74	0.06	0.82
CMJ (cm)	40.93 ± 5.80	41.29 ± 6.34	0.03	0.88
MED_THR (m)	4.88 ± 0.47	5.35 ± 0.63	5.12	0.03 *
POL_BACK (s)	10.51 ± 1.94	9.09 ± 1.52	4.83	0.04 *

Legend: PLANK—The plank hold; SQJ—squat jump; CMJ—counter movement jump; MED_THR—Medicine ball throw from a seated position; POL_BACK—Backward agility test; *—$p < 0.05$

Results presented in Table 2 suggest differences between the two groups in three motor ability assessment tests. A significant difference was found in the PLANK test, assessing static strength and trunk stability (PLANK—F = 10.40, $p = 0.00$). The average results in the PLANK test show that the group performing the shot with one hand from the spot had a significantly worse result, indicating weaker trunk strength and stability. A statistically significant difference was also found in the MED_THR test, which assesses upper limb strength (F = 5.12, $p = 0.03$). Looking at the average result, it can be concluded that G1 achieved a worse result, indicating that the strength of the upper limbs is at a lower level compared to G2. The difference between the two groups was also identified in the POL_BACK test (F = 4.83, $p = 0.04$), which assesses the coordination of the participants. Additionally, participants who performed jump shots achieved better results compared to the group that performed shots with one hand from the spot.

4. Discussion

The results of the kinematic parameters for both shooting techniques investigated in this study clearly and unequivocally demonstrate that the jump shot and shot with one hand from the spot are two distinct shooting techniques. This confirms the first hypothesis that the kinematic pattern of these two shooting techniques significantly differs (F = 12.86, $p = 0.00$).

The kinematic parameters of both shooting techniques were observed through ten variables, among which nine showed a statistically significant difference between the two shooting techniques. Despite the differences, these two shooting techniques also share some similarities. Apart from their common purpose, which is shooting the ball into the basket from a greater distance, similarities can be found in certain parameters.

The greatest similarities are found in the final phase of shooting, after lifting the ball toward the basket and before the ball's impact on the floor. The final phase of both shooting techniques also shares many similarities, as the concluding action on the ball originates from the hand. The speed of the hand segment itself does not show a significant difference. Additionally, according to kinesiological analysis, the last contact, including the rotation and the final impulse of force on the ball, is provided by the index and middle fingers imitating the "whip" motion, imparting sufficient force and precision for a successful shot. In this way, forces projected from contact with the floor and the strength of the musculature, especially the lower extremities, are more purposefully transferred to the ball. Consequently, although the entry angle of the ball statistically differs significantly, the values differ by approximately $\approx 2^{\circ}$. This, along with the release angle, is one of the significant parameters influencing the success of each individual shot [22,23].

The differences between these two shooting techniques are most evident in the preparation phase and all the actions that precede the actual release of the ball. The angle of the hand joint in the preparation phase for shooting significantly differs primarily due to the different body positions. The body position plays a crucial role because, in this phase, the shooter's body is positioned optimally for executing one of the two mentioned types of shooting. It is essential to place the body in a position where it can effectively utilize forces and levers for

the shot [24]. For example, a larger angle in the hand joint is recorded when shooting with a shot with one hand from the spot (Hand_rec—60.62 ± 15.43 vs. 51.90 ± 16.65), explainable by way of holding the ball on the chest. While performing the shoot with one hand from the spot, the player must generate the appropriate amount of force on the ball so as to reach the basket. Therefore, a player must position the ball closer to the body, consequently changing the angle of the wrist. Concerning the jump shot, the starting position is a triple threat stance—an offensive stance.

An interesting comparison arises from the speed of the upper limb segments, as well as the overall duration of the entire shot. A significant difference in speed in favor of the jump shot (jump shot = 0.87 s; shot with one hand from the spot = 0.97 s) was obtained even though higher speeds of the hand, forearm, and upperarm segments were recorded in the shot with one hand from the spot (Hand_V—4.94 ± 0.88 vs. 4.73 ± 0.71; Forearm_V—5.44 ± 0.79 vs. 4.73 ± 0.71; Upperarm_V—2.71 ± 0.38 vs. 2.46 ± 0.34). This might be due to the fact that in the jump shot, all activities must be synchronized in one biomechanical chain, making the jump shot a faster shooting technique. Here, the segments of the upper limbs alone exhibit significantly higher speed. The reason for this in our study might be the fact that the participants are still inadequately trained and formed, and such variations can be expected as a result of compensation. Namely, if the preliminary actions and preparations for performing either of the two shooting techniques, especially the jump shot, were not performed correctly, the deficiency in the initial phase (slower preparation) would be compensated for in the final phase, during the release. This results in a situation where the speed of the forearm and upperarm segments when the ball leaves the hand shows higher values in participants performing the shot with one hand from the spot.

Moreover, female basketball players also use the shot with one hand from the spot in their shooting repertoire from greater distances because of the lower level of certain abilities compared to elite male basketball players. In their study conducted on 10 elite senior female basketball players, Elliott and White [25] compared differences in some parameters between shooting for two and three points (the publication uses the term "jump shot"). In three-point shots, a statistically significant difference in the angles of the shoulder and hand was recorded in that study, which fully aligns with our results for the group of participants performing the one-handed chest shot from the spot. Okazaki and Rodacki [7], in their research, citing results obtained by Elliott [26], state that in female basketball players, precisely due to reduced force-generating capacity, during preparation, and later during the actual shot execution, there is greater body movement toward the basket and greater amplitudes in the measured angles in the shoulder joint. All these parameters indicate that the shot used in women's basketball more closely resembles the shot with one hand from the spot than the jump shot, even though it is in the literature referred to as a "jump shot".

In studies of a similar nature conducted on a former member of the U-20 national selection of Croatia [8,9], statistically significant differences (F = 184.69; p = 0.00) were also obtained between kinematic parameters in performing the jump shot and the shot with one hand from the spot. A statistically significant difference was found in the angle of the wrist (F = 66.84; p = 0.00). At the shot time, the variable analyzed was the angle in the shoulder joint of the shooting arm (F = 10.08; p = 0.01), horizontal movement of the player, or the length of the jump (10.22; p = 0.00), and a statistically significant difference was determined. In contrast to this study, there was no statistically significant difference in the variable of the ball entry angle into the basket (F = 3.27; p = 0.08).

One of the most characteristic parameters concerning the difference between the two types of shooting is horizontal movement (jump). As already mentioned, in the shot with one hand from the spot there is a much more pronounced forward movement of the body at the moment of shooting. This is confirmed by all relevant sources [15,27,28], including our research. The reason for this is the body position, ball holding, and the reduced ability of younger and undeveloped players who, extend their bodies toward the basket. For the same reason, the angle in the shoulder joint has a lower value during the shot with one

hand from the spot, where the arm extends forward, toward the basket, compared to the jump shot where the arm is raised more vertically.

Very similar results were also obtained by Svoboda [28], who demonstrated differences in some kinematic parameters between the jump shot and the shot with one hand from the spot in a sample of five cadet basketball players. The measured kinematic parameters showed a significant difference between the two groups in variables such as forward body movement (F = 1.60, p = 0.02) and the shoulder angle at ball release (F = 2.15; p = 0.00). The results align, and this is contributed to by the fact that the sample of participants is very similar since they were cadet players who, during the research period, were competing in the same category as the sample in this study. It can be concluded that players at this age and stage of biological and basketball development use both shooting techniques, which can be clearly defined, as well as the differences between them.

The results of the motor ability assessment tests in this study suggest statistically significant differences between the two groups of participants and confirm the second hypothesis that there is a statistically significant difference between the group of participants performing the jump shot and the group performing the shot with one hand from the spot (F = 5.51, p = 0.00).

So, the obtained results are in line with the expectations and assumptions set before the research. As mentioned earlier, the jump shot is a complex motor movement that, for correct and successful execution, must satisfy a certain number of conditions and parameters. One of them is the development of motor abilities at a certain level. The assumption was that selected motor abilities manifest through locomotor system movements crucial for executing the jump shot, especially correctly, purposefully, and efficiently. There is a limited number of research papers examining the relationships between motor abilities and the performance of a specific motor task.

In further analysis, it is essential to establish that certain motor abilities are significantly higher in the group performing the jump shot compared to the group of participants executing the shot with one hand from the spot (PLANK—125.84 ± 39.18 vs. 172.73 ± 39.07; MED_THR—4.88 ± 0.47 vs. 5.35 ± 0.63; POL_BAC—10.51 ± 1.94 vs. 9.09 ± 1.52). This inevitably leads to the conclusion that these abilities are a crucial factor determining the difference in execution, more precisely, the possibility of performing the jump shot as a technical element. A higher level of abilities in these variables means that, in this specific case, young players can execute the jump shot and, if not already, can start training and incorporating it into their repertoire for attempting shots from greater distances. On the contrary, basketball players who achieve lower results in these variables are not yet capable of performing the jump shot. It would be better for them and their further development to continue using the shot with one hand from the spot. Additionally, this information is valuable for coaches in terms of assisting in planning and programming the training process [29].

The backward agility test assesses coordination, a capability that significantly influences the execution of various movements and tasks, particularly shooting at the basket, especially from greater distances. As mentioned earlier, the jump shot is a highly complex motor skill, leading to the conclusion that a high level of coordination is necessary for its execution. The results of this research indicate that participants with a higher level of coordination were able to perform the jump shot, while those with a lower level of this ability used the shot with one hand from the spot. Performing the jump shot requires timely synchronization of the entire kinematic and biomechanical chain of movement.

During the preparation phase of the shot, a basketball player must generate significant force in the lower extremities, and this force must be transmitted through the torso to the distal joints of the wrists and finger joints to execute a throw that meets the criteria of a correct and efficient shot at the basket. Besides force transmission, generating significant force also affects the height of the jump during shooting. The trunk must remain stable during this phase to transfer the force generated in the lower extremities to the upper extremities [30]. In this study, trunk strength was assessed using the plank test. Apart

from force transmission to the upper extremities, trunk stability also plays a significant role in situational conditions during the execution of the jump shot, especially when this element is performed near an opposing player, i.e., in direct contact with them. As already mentioned, one of the greatest advantages of the jump shot over other types of shooting is rising above the defensive player [5]. If this elevation does not eliminate contact with the defensive player, further stability of the center of gravity and minimizing body oscillation undoubtedly enhance the performance of the jump shot. Considering the research results, the difference obtained between the two groups of participants unquestionably favors the group performing the jump shot. Specifically, the group performing the jump shot demonstrated a higher level of static strength in the trunk muscles compared to the group executing the one-handed chest shot from a standstill. This leads to the conclusion that static trunk strength is one of the factors determining the difference in the ability to perform the jump shot compared to the shot with one hand from the spot.

Continuing from the above, as already mentioned, the group performing the jump shot achieved better results in the MED_THR test, which assesses the explosive strength of the throwing type. These data were expected, considering that explosive strength, especially of the throwing type, is closely related to the execution of any type of shooting in basketball, particularly the jump shot. In basketball, almost all these types of movements are used, making explosive strength a significant factor contributing to success in basketball. Jumps and sprints are indispensable segments of basketball that are repeated many times during a game. However, the focus of this research is shooting, which falls under activities of the throwing type.

Limitations

Differences in kinematic parameters were observed between the group using the jump shot and the group employing the shot with one hand from the spot technique. Similarly, the assumption was that the groups would differ in motor skills, considering that they are a limiting factor for performing the jump shot. To further clarify the relationship between shooting technique and motor skills, specifically the impact of motor skills on the ability to perform each technique, it would be desirable to conduct a regression analysis. However, the number of participants in this study was not large enough to justify the implementation of a regression analysis. As a continuation of this research, it would certainly be desirable to expand knowledge about the direct impact of motor skills on specific kinematic parameters during shooting.

5. Conclusions

When categorizing participants into groups, experts divided them into two groups based on their assessment that the two shooting techniques differ. One hypothesis was that these two shooting techniques indeed differ, which was ultimately confirmed through kinematic analysis. The second assumption was that participants using two different shooting methods also differ in their level of motor skills. This assumption was also confirmed, with the expected higher level of motor skills observed in players using the jump shot technique. Therefore, players in younger age categories who have certain motor skills (leg strength, arm strength, torso stability, coordination) developed at an optimal level can perform a jump shot. While for those who do not have it, it is recommended that they perform a shot with one hand from the spot. Although the duration of certain phases (speed of the upper body segments) during the execution of the shot with one hand from the spot is short, the jump shot, looking at the entire movement structure, is even shorter. As a transitional technique to prevent the automation of the wrong movement, the shot with one hand from the spot is a good technique during the process of learning basketball.

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